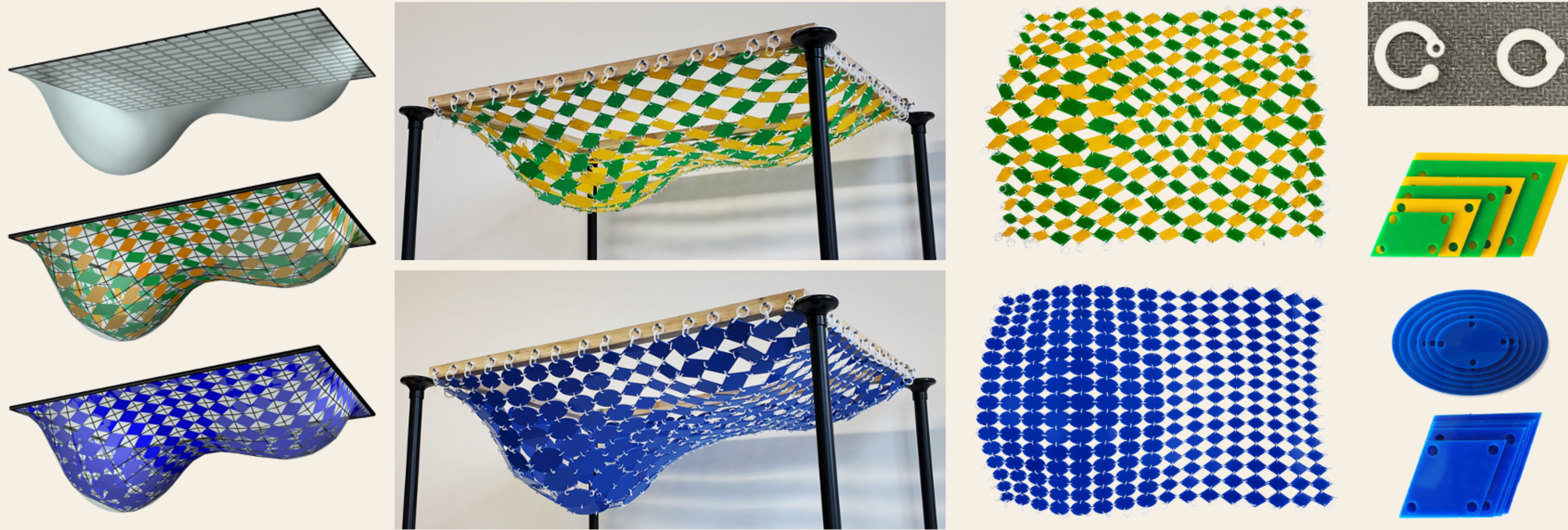




From similar elements to freeform surfaces

Analysis by isogonal nets on a quad mesh, freeform surface patterns are generated using simple similar elements. Local control of element shapes enables diverse global layouts while preserving a fabrication-friendly structure.

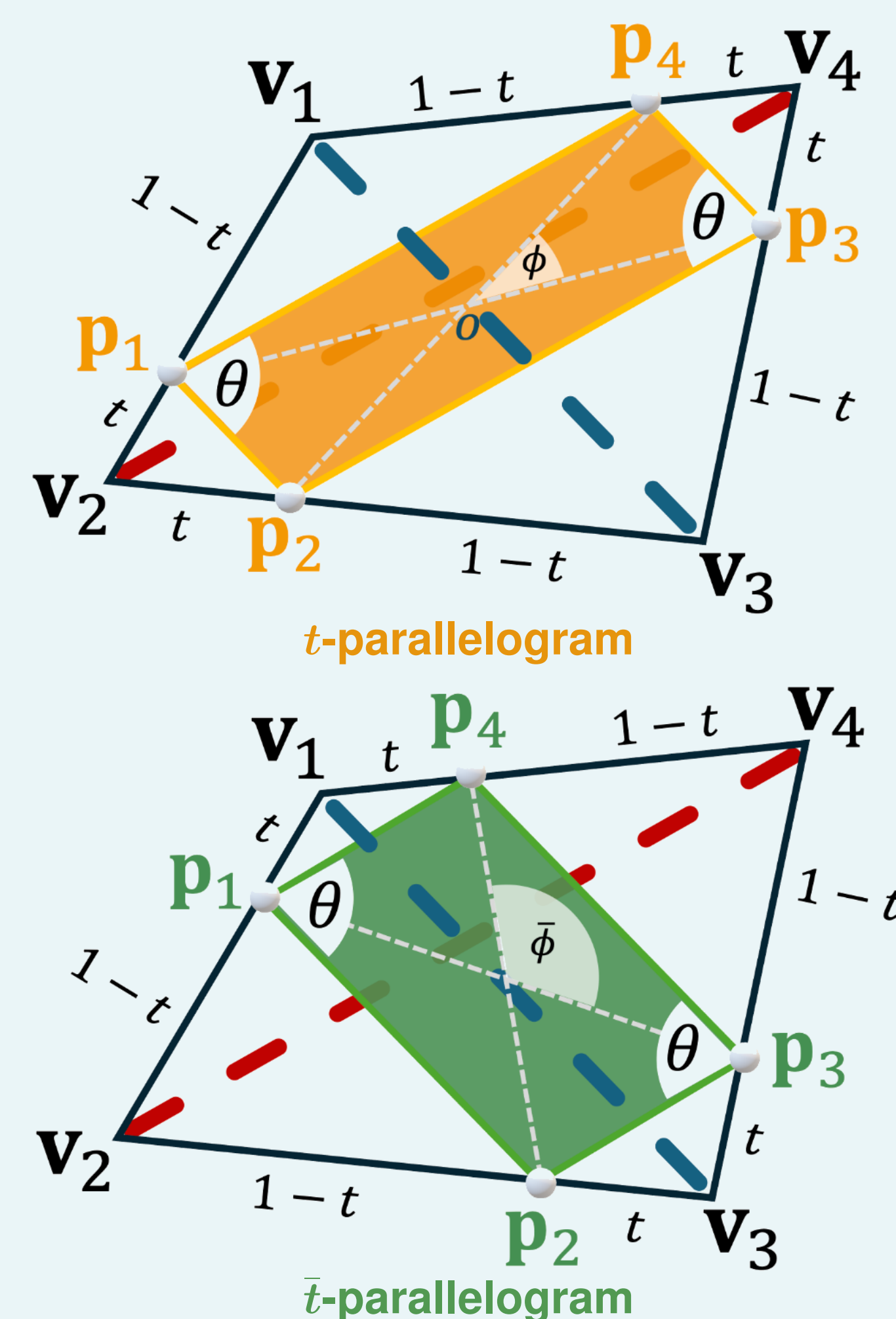
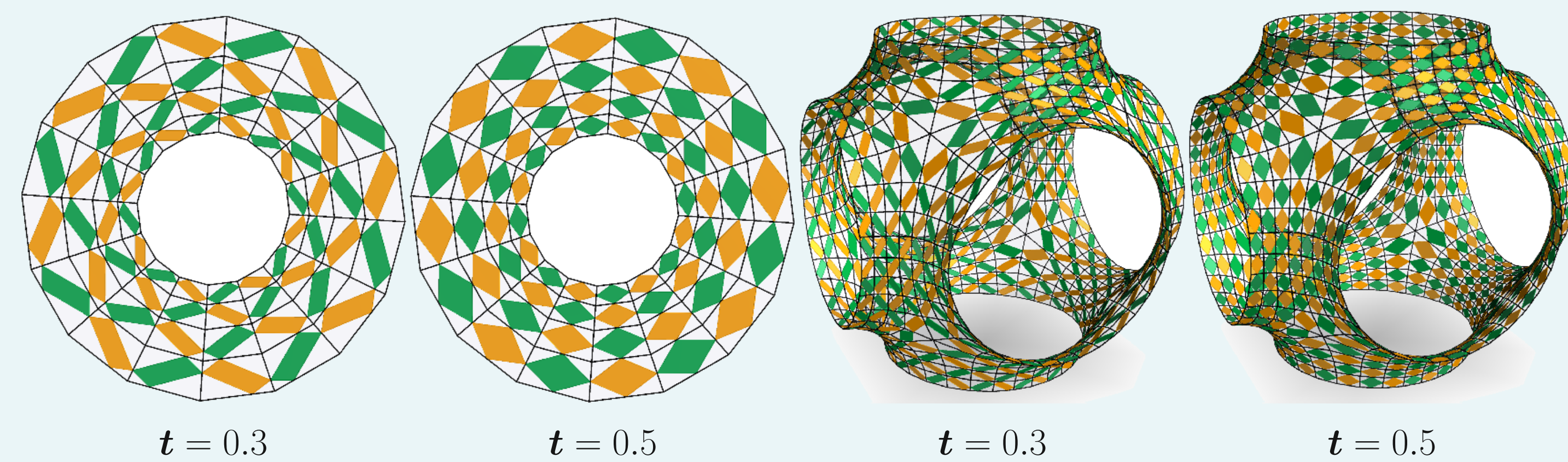


Checkerboard Pattern

For each quad face (v_1, v_2, v_3, v_4), gliding points controlled by one parameter t are placed on edges:

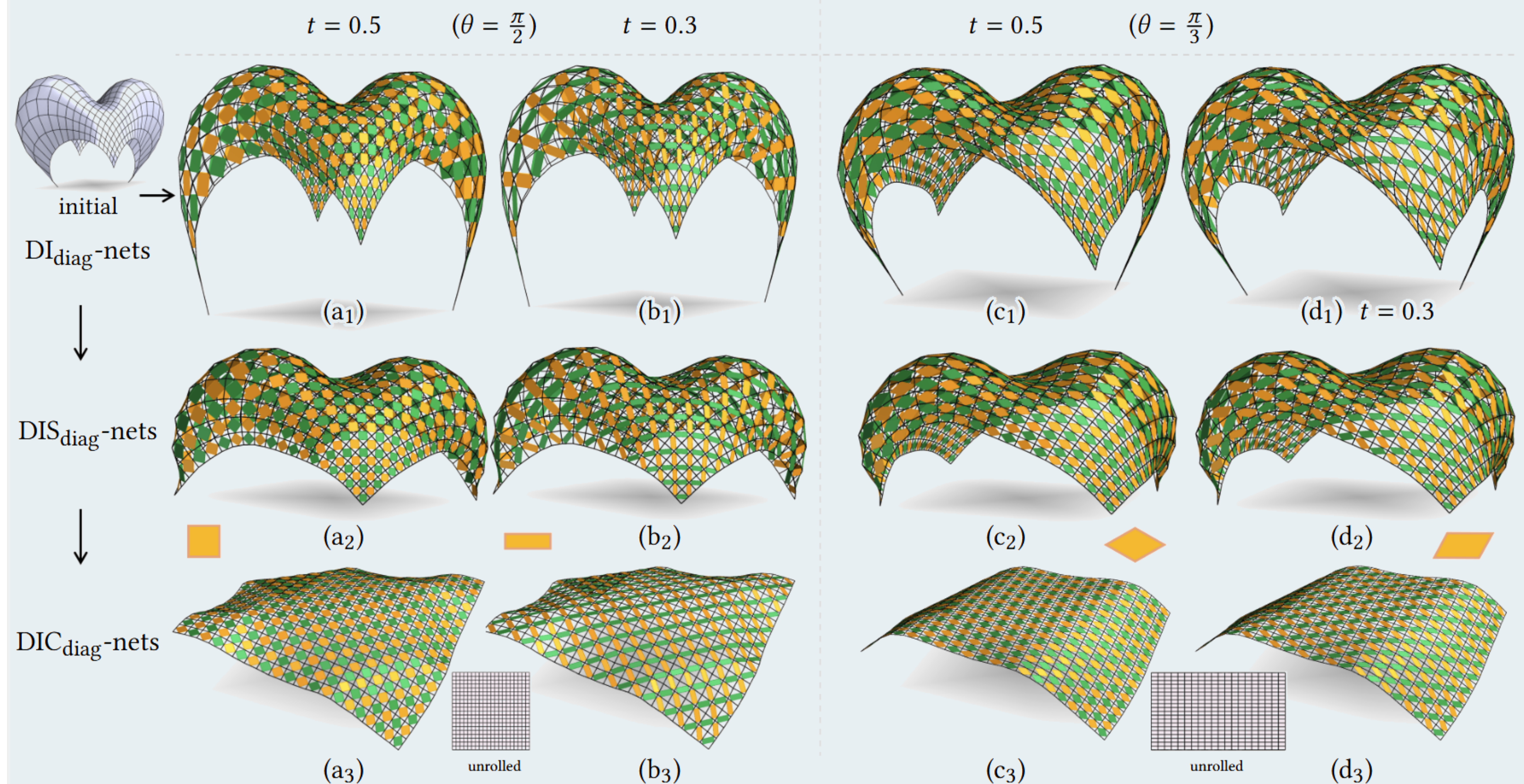
$$p_i = tV_i + (1-t)V_{i+1}, t \in [0, 1], \bar{t} = 1 - t.$$

Connecting these gliding points forms a t -parallelogram, whose shape is characterized by the interior angles θ and ϕ . Replacing t with $\bar{t}$ gives the corresponding $\bar{t}$ -parallelogram. As a result, extending the construction over the quad mesh produces checkerboard patterns.

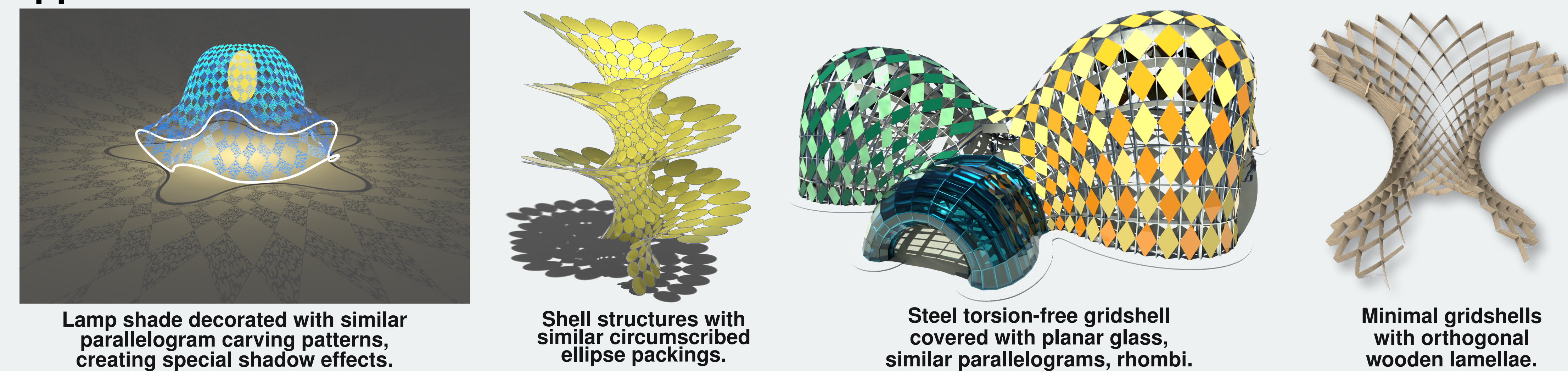


Optimization and Computational Design

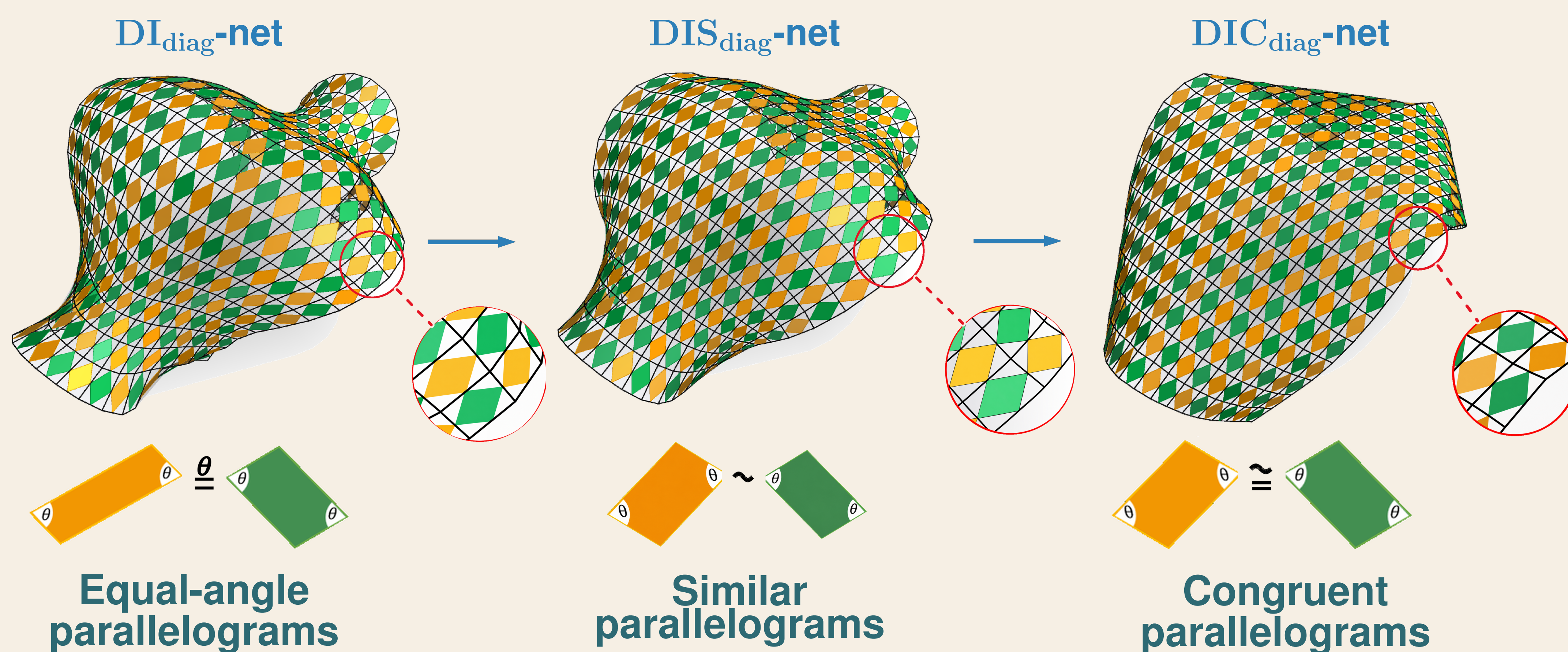
The initial mesh is progressively optimized into three classes of discrete isogonal nets:



The design flexibility of **DISdiag-nets** further enables a variety of computational design applications:



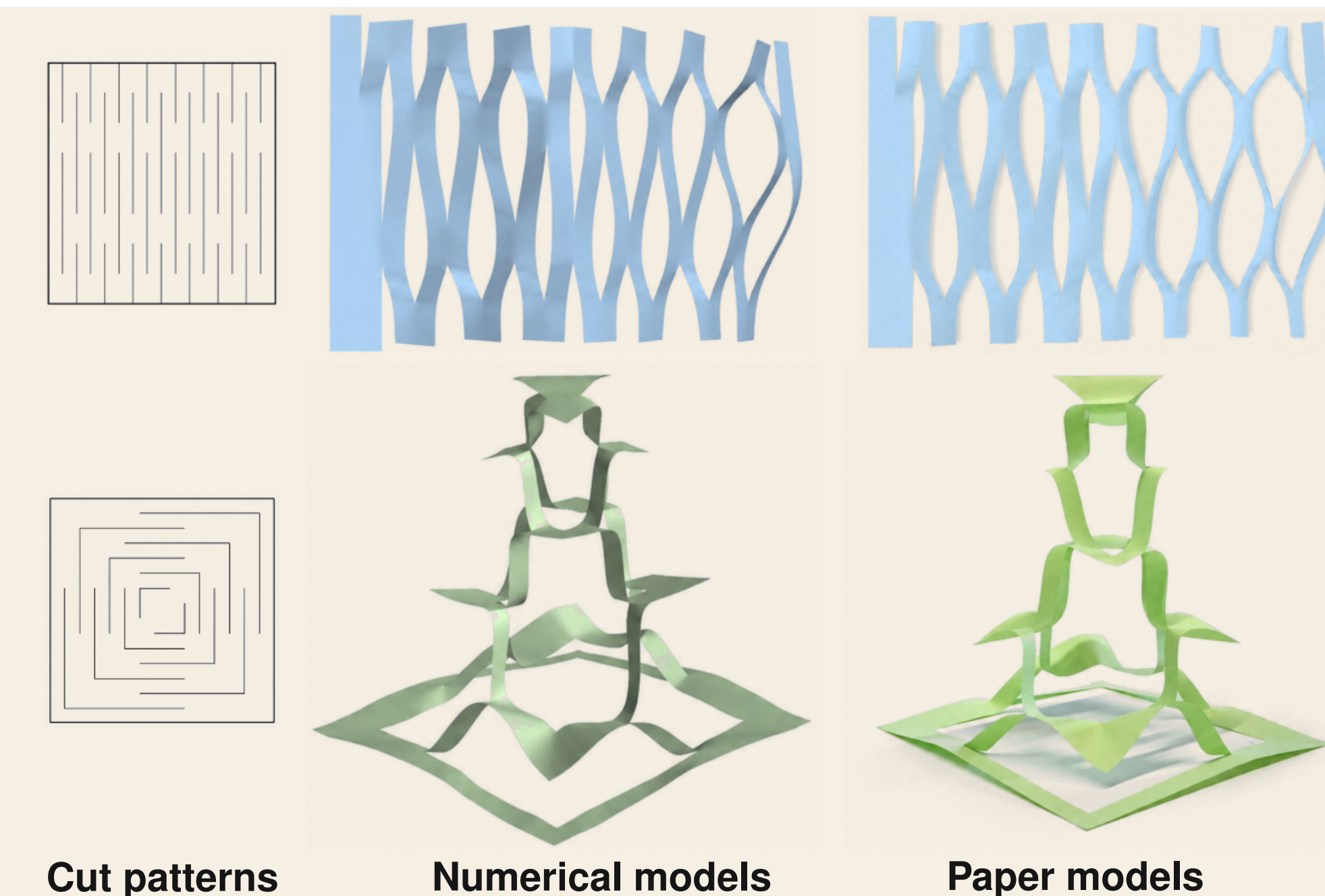
Definitions



Fabrication

Interactive editing preserves the **DICdiag-net** and local isometry to rectangular grids, enabling seam-based paper-cut fabrication.

As the teaser shown, the same rectangular-boundary freeform surface is tiled with two **DISdiag-net** patterns: yellow-green similar parallelograms at $t = 0.412485$, and similar rhombi with blue circumscribed ellipses at $t = 0.5$. Laser-cut prototypes reveal panel similarity.



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